

- Q.27 Write short note on testing and calibration. (CO3)
 Q.28 Describe solid state switches. (CO5)
 Q.29 What is the use of Karnaugh maps? (CO6)
 Q.30 Write short note on sequential logic. (CO6)
 Q.31 Write some applications of microcontrollers. (CO7)
 Q.32 How do you select a PLC? (CO10)
 Q.33 Differentiate between hydraulic and pneumatic systems. (CO4)
 Q.34 Write a short note on mnemonics. (CO8)
 Q.35 Give some applications of mechatronics. (CO1)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Define actuators. Explain any two hydraulics actuation systems. (CO4)
 Q.37 What is a stepper motor? Explain its construction and working with the help of neat sketch. (CO5)
 Q.38 Define programmable logic controllers. Explain its basic structure and applications. (Co10)

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6th Sem / Mech. Subject:- Mechatronics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A servo motor is a typical example of _____ (CO5)
 a) Electronics system b) Mechanical system
 c) Computer system d) Mechatronics system
- Q.2 The main function of Actuator is _____ (CO4)
 a) To produce motion
 b) Detect input
 c) Detect output
 d) Detect the state of the system
- Q.3 The main mechanical components of a servo motor are _____ (CO5)
 a) Only rotor b) Only stator
 c) Both stator and rotar d) None of these
- Q.4 Analog to digital conversion in a measurement system takes place inside _____ (Co1)
 a) Transducer b) Signal processor
 c) Display d) Led

- Q.5 What is the full form of LVDT with respect to displacement transducer? (CO2)
- Linear variable differential temperature
 - linear variable differential transformer
 - Liquid visible differential transformer
 - Liquified visible differential transformer
- Q.6 Which is the most common load cell used in the force sensors? (CO2)
- Hydraulic load cells
 - Strain-gauge based load cells
 - Pneumatic load cells
 - Capacitive load cells
- Q.7 The force developed in hydraulic systems is high due to _____. (CO4)
- High pressure
 - More oil
 - Less pressure
 - Less oil
- Q.8 Timer is used in PLC to ensure that the program, scan is completed in a timely. (CO10)
- Watchdog
 - Off delay
 - Retentive
 - on delay
- Q.9 PLC operates on _____ signals. (CO10)
- Analog
 - Digital
 - Impulse
 - Frequency
- Q.10 What is full form of CPR with respect to motor movement? (CO5)
- Clocks per rotation
 - Counts per revolution
 - Counts per rotation
 - Clocks per revolution

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What are the transducers? (CO2)
- Q.12 Write principle of working of DC motor (CO5)
- Q.13 Give disadvantages of mechatronics. (CO1)
- Q.14 Give an example of closed loop control system. (CO1)
- Q.15 What are the main types of sensors? (CO2)
- Q.16 Define liquid flow sensor. (CO2)
- Q.17 What is a pneumatic system? (CO3)
- Q.18 Define solenoids. (CO1)
- Q.19 Define logic gate (CO6)
- Q.20 What is the use of shift registers? (CO8)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define mechatronics. Give its advantages. (CO1)
- Q.22 Explain microprocessor based controllers. (CO7)
- Q.23 Describe motion sensor. (CO2)
- Q.24 Explain temperature sensor with examples. (CO2)
- Q.25 Write short note on magnetic recording. (CO3)
- Q.26 Explain rotary actuators. (CO4)